

Hoang (Bolton) Tran, Ph.D.

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Summary

Computational scientist with experience in mathematical modeling, physics simulation and machine learning.
Chemical engineer with expertise in electrochemistry, catalysis, and process modeling.

Skills

- Molecular Dynamics: LAMMPS, GROMACS, MACE, DeePMD
- Quantum Chemistry: Gaussian, VASP, JDFTx, NWChem
- Cheminformatics: RDKit, Open Babel, Pymatgen, Materials Project, Open Catalyst, PubChemQC.
- Machine Learning: Scikit-learn, TensorFlow, PyTorch
- Programming: Python, C++, MATLAB, VBA, SQL, HTML/CSS, Git, HPC (PBS, Slurm)
- Process Modeling: Aspen Plus/HYSYS, COMSOL

Work Experience

Thermodynamics Sr. Research Investigator—*The Chemours Company* Dec 2025 – Present

- Develop fluid and process models to support the development of chemicals for thermal solutions.
- Streamline first-principles and data science workflow for the discovery of new refrigerant molecules.

Postdoctoral Research Fellow—*University of Michigan, Ann Arbor, MI* Sept 2023 – Nov 2025

- Investigate materials for electrochemical applications using quantum chemistry and machine learning.
- Simulate X-ray absorption spectra of supersaturated vanadium electrolytes using machine learning force-fields.
- Decode physical descriptors for electrolytic anion adsorption using quantum chemistry and regression models.
- Elucidate electrochemical reaction network of CO₂ electro-reduction using a quantum-based microkinetic model.

Doctoral Research Assistant—*Pennsylvania State University, University Park, PA* Sept 2018 – Aug 2023

- Advanced fundamental understanding of electrochemistry and catalysis through molecular dynamics.
- Investigated water and ions at the solid/liquid interfaces by developing a polarizable force-field.
- Refined classical force-fields for aqueous electrolyte to match experimental osmotic pressures.
- Elucidated organic solvents' impact on biomass reaction kinetics through multi-scale modeling.

Quality Assurance Associate—*Avid Radiopharmaceuticals, Philadelphia, PA* Nov 2016 – Aug 2018

- Managed the quality control for the production of Amyvid, a radiotracer drug for Alzheimer's disease.
- Designed a Quality Assurance Scorecard used to evaluate performance of 20+ manufacturing contractors.
- Developed a semi-automated central database to track contractors' production status and inventories.

Analytical Chemistry Intern—*Zeolyst International, Conshohocken, PA* May 2014 – Sept 2014

- Analyzed zeolites for the Selective Catalytic Reduction reaction in catalytic converters.
- Assessed catalytic performance of zeolites through gaseous infrared spectroscopy.
- Monitored gas flow rates, temperatures and pressures of two micro-reactors daily.

Education

Pennsylvania State University, Ph.D. in Chemical Engineering Sept 2018 – Aug 2023

Drexel University, B.S. in Chemical Engineering (GPA: 3.83) Sept 2012 – June 2017

Publications

1. **Tran, B.**, Goldsmith, B. R., “Predicting competitive anion electrosorption on late transition metals”, *Chem. Sci.* **16**, 17325–17333 (2025)
2. Mathanker, A., Sharma, G., **Tran, B.**, Singh, N., Goldsmith, B. R., “Effect of Ions on the Aqueous-Phase Adsorption of Small Aromatic Organics on Silver”, *J. Phys. Chem. C* **129**, 13433–13444 (2025)
3. Sweeney, D. M., **Tran, B.**, Goldsmith, B. R., “A grand canonical study of the potential dependence of nitrate adsorption and dissociation across metals and dilute alloys”, *Commun. Chem.* **8**, 182 (2025)
4. Long, E., **Tran, B.**, Milner, S. T., “Tuning partial charges of alkyl alcohols to improve simulated fluid properties”, *J. Chem. Phys.* **162** (2025)
5. **Tran, B.**, Janik, M. J., Milner, S. T., “Hydration-Shell Solvation and Screening Govern Alkali Cation Concentrations at Electrochemical Interfaces”, *J. Phys. Chem. C* **128**, 20559–20568 (2024)
6. Mathanker, A., Halarnkar, S., **Tran, B.**, Singh, N., Goldsmith, B. R., “Synergistic effects in organic mixtures for enhanced catalytic hydrogenation and hydrodeoxygenation”, *Chem Catal.*, 101135 (2024)
7. **Tran, B.**, Goldsmith, B. R., “Theoretical Investigation of the Potential-Dependent CO Adsorption on Copper Electrodes”, *J. Phys. Chem. Lett.* **15**, 6538–6543 (2024)
8. Wong, A., **Tran, B.**, Agrawal, N., Goldsmith, B. R., Janik, M. J., “Sensitivity Analysis of Electrochemical Double Layer Approximations on Electrokinetic Predictions: Case Study for CO Reduction on Copper”, *J. Phys. Chem. C* **128**, 10837–10847 (2024)
9. **Tran, B.**, Zhou, Y., Janik, M. J., Milner, S. T., “Negative Dielectric Constant of Water at a Metal Interface”, *Phys. Rev. Lett.* **131**, 248001 (2023)
10. Ostervold, L., Daneshpour, R., Facchinei, M., **Tran, B.**, Wetherington, M., Alexopoulos, K., Greenlee, L., Janik, M. J., “Identifying the Local Atomic Environment of the “Cu₃+” State in Alkaline Electrochemical Systems”, *ACS Appl. Mater. Interfaces* **15**, 27878–27892 (2023)
11. **Tran, B.**, Milner, S. T., Janik, M. J., “Kinetics of Acid-Catalyzed Dehydration of Alcohols in Mixed Solvent Modeled by Multiscale DFT/MD”, *ACS Catal.* **12**, 13193–13206 (2022)
12. **Tran, B.**, Cai, Y., Janik, M. J., Milner, S. T., “Hydrogen Bond Thermodynamics in Aqueous Acid Solutions: A Combined DFT and Classical Force-Field Approach”, *J. Phys. Chem. A* **126**, 7382–7398 (2022)
13. Edley, M. E., Opasanont, B., Conley, J. T., **Tran, H.**, Smolin, S. Y., Li, S., Dillon, A. D., Fafarman, A. T., Baxter, J. B., “Solution processed CuSbS₂ films for solar cell applications”, *Thin Solid Films* **646**, 180–189 (2018)

Conference Presentations

- BASF NA Research Forum 2025 | Wyandotte, MI | poster presentation.
- North American Catalysis Society 2025 | Atlanta, GA | oral & poster presentations.
- American Institute of Chemical Engineering Meeting 2024 | San Diego, CA | oral & poster presentations.
- American Chemical Society Fall Meeting 2024 | Denver, CO | oral & poster presentations.
- Gordon Research Seminar/Conference 2024 | New London, NH | oral (invited) & poster presentations.
- Michigan Catalysis Society 2024 | Ann Arbor, MI | oral presentation.
- American Chemical Society Spring Meeting 2023 | Indianapolis, IN | oral presentation.
- American Physical Society March Meeting 2022 | Chicago, IL | oral presentation.

Awards

- Sustainability challenge–1st place | BASF NA Research Forum 2025
- University of Michigan postdoctoral travel award | American Chemical Society Fall Meeting 2024
- ACS CATL ChemCatBio travel award | American Chemical Society Spring Meeting 2023
- Poster presentation–1st place | Pittsburg-Cleveland Catalysis Society 2022
- FGSA travel award | American Physical Society March Meeting 2022
- Departmental best qualifying exam | Pennsylvania State University 2019